QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION MONTH 2026
FIFTH SEMESTER
DSC- MG5DSCBIF301- Systems and Synthetic Biology
(2024 ADMISSION ONWARDS)

Duration: 2hrs **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Define system biology. [K] / [CO1]

2. Describe feedback loops. [U] / [CO2]

3. Interpret Regulatory Circuits. [An] / [CO3]

4.	Write a note on flux analysis.	[A] / [CO4]
5.	Describe synthetic biology.	[U] / [CO5]
6.	Interpret Robustness.	[An] / [CO6]
7.	Explain Principles of Bacterial Photography.	[U] / [CO7]

[2x5 = 10]

Part B

Short Essay Questions

Answer any 6 questions

Each question carries **5** marks

8.	Memorize history of system biology.	[K] / [CO1]
9.	Describe gene control and genetic switches.	[U] / [CO2]
10.	Interpret Protein-Protein Interaction (PPI) Networks.	[An] / [CO3]
11.	Describe definition and Goals of synthetic biology.	[U] / [CO5]
12.	Evaluate Synthetic Biology in Medicine.	[An] / [CO8]
13.	Explain E-Cell Project.	[An] / [CO6]

14.	Analyze Green Fluorescent Protein (GFP) Project.	[An]/[CO8]
15.	Discover Risks and Benefits of research in Synthetic Biology.	[U]/[CO5]
16	Explain Flux Analysis in detail.	[A]/[CO4]

[6x5 = 30]

Part C

Long Essay Questions

Answer any 3 questions

Each question carries **10** marks

17.	Explain Environmental Applications of Systems and Synthetic Biology	[U] / [CO5]
18.	Explain the biochemical, genetic and the systems paradigm.	[U] / [CO2]
19.	Explain Protein-Protein Interaction (PPI) Networks.	[An] / [CO3]
20.	Explain Bacterial Camera Project.	[A] / [CO3]

[10x3 = 30]